

VOLTRON STATIC — a research protocol

STATUS: DRAFT protocol. NOT A PRODUCT.

One of three documents, and the only one that is not a specification.

Document	Answers	Is it a product?
DRAFT_ORGANISM_AGENT_SPEC.md	What makes a composition the same continuing entity	Yes — this is the thing
DRIFT_EXTENSION.md	Does the entity still mean what it said	An extension of it
VOLTRON_STATIC.md (this)	How would we <i>measure</i> that	No — a laboratory protocol

STATIC freezes the base system in order to measure it. **It is not a revised identity policy and not a deployment posture.** Earlier drafts embedded a full copy of the specification; this one references it instead, so the two cannot drift out of sync.

1. Why a protocol at all

The base specification permits the brain to be replaced. The drift extension shows that unchanged commitments do not guarantee unchanged interpretation. Put together, that leaves a measurement problem: **when behaviour moves, too many things could have caused it to attribute the change to any one of them.**

STATIC's answer is to remove the removable causes, control what can be controlled, declare the rest, and measure what remains. It buys inference at the cost of realism — which is what a laboratory is for.

The strongest claim STATIC supports:

A controlled apparatus for measuring how memory, changing situations, and self-referential precedent affect behavioural continuity under a fixed composition.

Not "we have isolated all drift." That claim is not available (§2.4).

2. The protocol

2.1 What STATIC is, and what it is not

STATIC is a constrained experimental posture. It is not a replacement for the base organism policy, and the two do not contradict each other.

The base spec (§6.5) holds that the brain is replaceable — *"a sharper mind is the same entity thinking better."* STATIC pins the organ set, so under STATIC a brain replacement becomes a commitment amendment. That is a deliberate suspension of §6.5 **for measurement**, in the way a laboratory holds a variable fixed without claiming the world does. Whether STATIC is also a sane deployment posture is a separate question this document does not answer.

The claim STATIC actually supports — and the strongest one available:

We have built a controlled apparatus for measuring how memory, changing situations, and self-referential precedent affect behavioural continuity under a fixed composition.

Not "we have isolated all drift." That claim is not available, for the reasons in §2.4.

2.2 What the freeze removes

Cause	Under STATIC	Note
Continued training / fine-tuning	Removed	Weights are static artefacts; there is no training loop. Learning happens in memory and context, never in the organ.
Training-time optimisation pressure	Removed	Requires a reward signal feeding back into the weights. With no training loop there is nothing for it to act on.
Inference-time social pressure (sycophancy)	NOT removed	⚠ A static model can become sycophantic without any weight update — through user feedback within a conversation, framing, or an evaluation objective it is responding to. Earlier drafts wrongly filed all sycophancy under training pressure. Only the training-time half is removable.
Runtime change	Removed by pinning	The §9 runtime measurement is pinned in the commitment set; a prompt or tool-manifest edit becomes a commitment amendment. Built and enforced.
Organ replacement	Removed by pinning	The full organ set is pinned. Built and enforced. Cost: §6.5's upgradeability, suspended.
Memory accumulation	Remains	Memory is the continuity substrate; removing it removes the identity.
Distributional shift	Remains	The world moves regardless.
Interpretation ratchet	Remains	An entity that reasons about borderline cases sets precedent for itself.

2.3 STATIC's rules, as verifier semantics

STATIC's pins are **named rules**, not prose. They engage only when the commitment set pins them — a lineage that pins nothing is not under STATIC — and they are enforced by the same verifier that runs V1–V6.

S1 — Organ pin. If the commitment set names an organ set, `R.organs` must equal it exactly, role for role. Under STATIC, changing *any* organ therefore requires a quorum-authorized commitment amendment.

S2 — Runtime pin. If the commitment set names a runtime measurement, `R.runtime` must equal it. A prompt or tool-manifest edit becomes a commitment amendment rather than an invisible change.

S3 — Commitment binding. The commitment-set artefact explicitly contains the pinned **organ set**, the **runtime measurement**, the **conscience organ hash**, and the **probes**. Because the commitments hash covers all four, none can be altered without the amendment quorum — and the conscience organ can be neither swapped for a compliant one nor unplugged while the probes sit anchored with nothing to run them.

2.4 Δ Experimental confounds — stated before the experiment, not discovered by it

A freeze of the composition is not a freeze of everything. These survive it, and an experiment that does not control or measure them will attribute their effects to the factors under test:

- **Sampling nondeterminism** — temperature, top-p, seed.
- **Numerical and environmental differences** — library, kernel, driver, hardware, batching, quantisation kernels. The same weights on different silicon are not the same function.
- **Context ordering and serialisation** — the same content, assembled in a different order or format, is a different input.
- **Retrieval implementation** — an unchanged index queried by changed retrieval code returns different context.
- **Tool and environment outputs** — the entity's inputs include the world's answers, which move.
- **Time-dependent external state** — anything dated, cached, or live.
- **Conversational framing and user interaction** — including the inference-time sycophancy above.
- **Context contamination and prompt injection** — hostile or accidental content entering the context window.

A specification that omits these does worse than ignore them: it guarantees the experiment will find a confounder and mistake it for a result. They are listed here so the harness must either control them or declare them uncontrolled.

2.5 The three factors — as interventions, not as causes

M and R are not causally independent, and this document does not pretend otherwise.

Memory exposure and precedent exposure are both forms of *context exposure*; precedent is a particular arrangement of memory rather than a separate mechanism. They are separable as **manipulations**, which is all a factorial design requires.

	Intervention	Definition
M	Autobiographical context	Supplied, with prior decisions removed — so M does not silently carry R
R	Self-referential precedent	Prior decisions supplied, autobiographical context held constant
D	Situation distribution	Varied independently of both

Interaction terms are expected, not anomalous. Because M and R share an underlying channel, additivity would be the surprising result. The harness reports interaction size; it does not treat its presence as a finding.

2.6 Canonical cell inputs — the operational definition

"Held constant" means nothing until it is a serialisation rule. A factor is only interpretable if two cells differ in **exactly** the intended way and in no other, so every cell is built under declared inclusion/exclusion rules, canonicalised, and **hashed** — making "these cells differ only in M" checkable rather than asserted.

Question	Rule
What counts as autobiographical context (M)?	Narrative log entries only, most recent first, capped at <code>memoryBudget</code>
Are summaries of prior decisions in M?	No. A summary of decisions is R in condensed form; both decision records <i>and</i> decision summaries are excluded from M
What does "M held constant" mean when R is toggled?	The autobiographical slice is computed independently of R and is byte-identical across the R toggle
What is in R?	Prior decisions only, most recent first, capped at <code>precedentWindow</code>
Is context order canonicalised?	Yes — recursive key sort, arrays keep meaningful order
Are budgets and decoding fixed?	<code>memoryBudget</code> and <code>precedentWindow</code> are declared inside the cell input, so any change alters the cell digest and cannot pass unnoticed. Seed, decoding parameters, runtime image and tokenisation must be pinned by the operator and recorded alongside
What about anything not listed?	Absent by default. Inclusion must be declared; nothing enters a context implicitly

Each condition records the hash of its cell inputs. **A sweep whose cells cannot be shown to differ only in the intended factor is not a factorial experiment**, whatever its numbers say.

2.7 The design and its null

A factorial sweep over $M \times D \times R$ — eight conditions, each run multiple times so stochastic spread is visible rather than assumed away.

The null, stated as a tolerance:

The baseline condition must show **no drift beyond a pre-registered tolerance**, and no spread across repeats beyond that tolerance.

Demanding *zero* drift would be either unsatisfiable or a false claim of determinism. Two honest routes to a small tolerance:

1. **Pin determinism** — seed, decoding parameters, hardware and runtime image, and input serialisation. Then the tolerance can approach zero legitimately.
2. **Accept stochasticity and measure distributions** — compare response distributions or behavioural scores across repeats, never exact strings, and pre-register the tolerance and the comparison method *before* running.

If the baseline exceeds tolerance, the harness reports failure and attributes nothing. **A sweep that cannot fail its own null is not an experiment.**

2.8 What this cannot tell you

The apparatus is not a result. The harness takes a `brain` — any function from situation and context to response. Against a real model the numbers say something about that model; against a synthetic brain they say only that the apparatus works. No output is evidence about real systems unless a real system produced it.

The freeze has a cost. It suspends exactly the upgradeability §6.5 was built around. That is the right price for a laboratory and not obviously right for an entity meant to operate for years.

The open question, unchanged: can a fixed composition be reasoned toward minimal drift — or is some irreducible drift the price of interpreting a changing world at all?

3. Status, and when to run it

Built and tested: `src/experiment.js` (canonical cell inputs, the factorial sweep, the tolerance null, the interaction test), the S1/S2/S3 pins in `src/verifier.js`, and `test/experiment.test.js` — which exercises the apparatus itself, including that a stochastic brain fails the null rather than producing numbers.

Never run against a real model. Every number this harness has produced came from a synthetic brain and means only that the apparatus works.

When to run it: after the base specification is real — anchored to an actual chain, over artefacts that actually exist. A protocol for measuring drift in a system that is not yet running measures nothing.

Appendix — revision history

rev 4 (2026-08-03). Rewritten as a standalone protocol that references the specification instead of embedding a copy of it, and relabelled explicitly as not-a-product. Sections renumbered from §12.6.x to §2.x. No substantive change to the protocol itself.

rev 3. Formalised what STATIC claimed to enforce: `restore` in the schema, S1/S2/S3 named as verifier rules, V3's genesis exception, canonical cell inputs with per-cell digests.

rev 2. Confounds stated before the experiment; M/D/R redefined as interventions rather than independent causes; tolerance-based null; training-time optimisation separated from inference-time sycophancy; STATIC stated as a posture rather than a policy.

rev 1. Base spec plus the frozen-baseline section.